

Swadesh Singh Rathore

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EDUCATION

Swami Keshvanand Institute of Technology (SKIT) <i>B.Tech Honours in Computer Science (Artificial Intelligence) – CGPA: 9.00/10</i>	Jaipur, Rajasthan Sep. 2023 – 2027
SBS Convent School <i>High School (PCM) – 78.60%</i>	Sikar, Rajasthan 2022
SBS Convent School <i>Secondary School (10th) – 81.20%</i>	Sikar, Rajasthan 2020

EXPERIENCE

Quantum Computing Intern <i>Defence Research and Development Organisation (DRDO) – SAG Lab</i>	May 2026 – July 2026 New Delhi, India
- Benchmarked quantum error mitigation methods on cryptanalytic circuits using Qiskit and IBM hardware tools. - Implemented Shor's and Grover's algorithms in Qiskit to simulate cryptographic key-search workloads on Aer. - Applied TREX readout mitigation and 9-qubit error correction codes on real IBM quantum hardware processors.	
Data Engineer Intern <i>Celebal Technology</i>	May 2026 – July 2026 Jaipur, Rajasthan
- Engineered SQL and Python data cleaning workflows, lifting dataset accuracy and pipeline efficiency by 30%. - Designed Azure ETL pipelines with ADF and Delta Lake, applying medallion architecture for clean data flows.	
Technical Intern <i>Modern Insulators Ltd.</i>	June 2025 – August 2025 Abu, Rajasthan
- Engineered backend web systems with MySQL, analyzing data pipelines to boost handling efficiency by 30%. - Designed FAQ module with structured SQL schema, cutting customer support response time by 35%.	

PROJECTS

Smart Fraud Detection Pipeline <i>ADF, Databricks, PySpark, ETL, SQL, Python, Medallion Architecture.</i>	LIVE
- Built Azure ELT pipelines using ADF, Databricks, PySpark, and Delta Lake for scalable data processing. - Implemented Medallion Architecture with SQL and Python for data cleaning, validation, and analytics readiness.	
QuantumBase <i>Python, Qiskit, IBM Quantum, QEC, QEM</i>	LIVE
- Implemented error correction techniques including Bit-Flip, Phase-Flip, and Shor's 9-qubit code using Qiskit. - Applied error mitigation methods such as Pauli Twirling, Dynamical Decoupling, TREX, ZNE, PEC, and PEA for NISQ systems.	
ResumeBuilder & ATS Checker <i>Flask, MySQL, HTML, CSS, JavaScript</i>	LIVE
- Built a Flask-based resume platform with MySQL, enhancing ATS keyword match accuracy by 25% for recruiters. - Designed optimized database schema cutting resume retrieval query response time by 40% across 100+ records.	

TECHNICAL SKILLS

Languages: Python, Java, SQL, SQL Advance
Cloud & Data Engineering: ADF, ADLS Gen2, Azure Databricks, Delta Lake, PySpark, Power BI, ETL & ELT
Quantum Computing: Qiskit, IBM Quantum, QEM, QEC, Cryptanalysis, Shor's & Grover's Algorithms
Data Science & Analytics: Statistical Analysis, EDA, Matplotlib, Seaborn, Pandas, NumPy, Scikit-Learn, ML
Web Development & Databases: Flask, MySQL, SQLite, HTML, CSS, Bootstrap, Figma
Soft Skills: Analytical Thinking, Problem Solving, Communication, Leadership, Teamwork, Adaptability

CERTIFICATIONS & ACHIEVEMENTS

Secured 100% scholarship in RCAT Rajasthan(2026), demonstrating strong academic excellence and merit.
Earned NPTEL Elite Certificate in Programming in Java (2025), showcasing strong core programming proficiency.
Earned NPTEL Silver Medal Certificate in Joy of computing in python (2026)
Preliminary Round Qualifier — Smart India Hackathon (SIH)(2024).
Student Coordinator, IIRS ISRO Outreach Program